The First Distinction

A Companion to Void and Form

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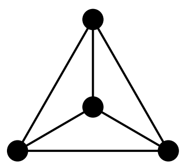

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Opening

Why is there anything at all, rather than nothing?

Since Leibniz, this question has accompanied every inquiry into origin. In modern cosmology it appears again: the closer the description of the universe moves toward its beginning, the clearer a limit becomes. Even an ever more exact history of the cosmos would still not explain why there is a cosmos at all.

The question already presupposes a distinction. *Something* and *nothing* must be in force as different possibilities. A nothing that could not be distinguished from something could not appear as nothing. The question of origin thus leads to a more general question: under what conditions is distinction possible at all?

The suspicion that difference, determination, and information belong together is not new. It appears in different forms in Aristotle, Hegel, Spencer Brown, Gregory Bateson, Claude Shannon, John Archibald Wheeler, and in relational ways of thinking in modern mathematics and logic. These names are not invoked as authorities. They mark only the direction to which this book belongs.

That is why the book does not begin with things, objects, or an ontology. Its starting point is the question of what must happen for anything at all to appear as different from something else. Its guiding working assumption is this: determination presupposes successful distinction.

If every description begins with a distinction and physical reality hangs on information, then before measurement, classification, and explanation there must already be a stable difference available. The book therefore asks for the minimal structure that carries a distinction: it must be stateable, it must not collapse, and it must be complete enough to make information possible.

Up to the formalization, the text speaks at the level of philosophical orientation and working assumptions. Only later do formal results enter. Physical readings are separated from those again. The formal core therefore does not answer Leibniz's question as a whole. It marks only the first point at which this condition can be formulated exactly and checked in intensional Martin-Löf type theory with Agda.

The path can be read as a sequence of questions. Why do origin, determination, physics, and information point toward distinction? What conditions must a successful distinction satisfy? In what language can those conditions be written exactly? What is the first such form? What already follows from it? What has thereby been shown, and what has not? And what happens when the same distinction is allowed to continue running in *Void* and *Form*? Each answer generates the next question.

Contents

Chapter 1

Why Distinction?

> "Why is there something rather than nothing? For nothing is simpler and easier than something."
>
> Gottfried Wilhelm Leibniz

The question of origin, determination, physics, and information approach the matter from four different sides. This chapter argues that they point to the same condition. It gathers not four unrelated claims, but four witnesses to one conjecture: that determination presupposes successful distinction.

The Question of Origin

In 1714 Leibniz formulated the question of origin in a form that remains sharp: why is there something rather than nothing? Nothing, Leibniz says, is simpler and easier than something. If the world could just as well have been empty, then precisely its not being empty calls for an account.

No measurement answers this question. One can trace the history of the cosmos back to the first physically describable state and still leave open why there is a cosmos whose history can be traced at all. The question reaches beyond everything data alone can decide.

But the question already carries one of its own conditions. To ask between *something* and *nothing*, both must already be in force as different. A nothing that could not be distinguished from something could not appear as nothing. The question of origin therefore already presupposes a distinction it does not itself explain.

That is why it leads immediately further. Not only: why is there something rather than nothing? But also: what must already hold for something to be distinguishable from nothing at all?

Determination

A brief answer to that last question is: it must be possible at all to set something off from something else. This condition concerns not only the contrast between something and nothing. It holds wherever something is determined as this and not that.

A simple case is enough.

A mark is this mark and not the blank place on the page. This sheet is this sheet and not that one. One side is this side and not the other. Wherever something is determinate, something else is excluded.

One cannot step behind holding-apart itself in search of a purer ground. Without this elementary difference there would be no determinate something. A sign that is not distinguished from what it refers to is no sign. A mark that does not stand out from the sheet on which it stands is no mark.

At first this describes only the simplest case. But its reach is clear: wherever something appears as this and not as that, something must already be distinguished from something else.

If that is right, the point should show itself most clearly where determination has been driven furthest: in physics. Does physics encounter here only a limit of our knowledge, or a condition for anything to appear as different at all?

The Physical Limit

If this condition is real, it must be visible in physics. There the attempt to grasp the beginning of the universe as a state of the world has been driven furthest.

Modern cosmology reconstructs the history of the universe backward. Galaxies move closer together, the universe becomes hotter and denser, atoms dissolve into plasma, nuclei break apart. This reconstruction is quantitative, tested, and in important regions very precise. Yet it does not reach its first state. The closer physics comes to the beginning, the thinner the data become. Light does not reach us from before a certain point. Energies climb beyond everything an experiment can reproduce. Space and time themselves lose the status of magnitudes that could simply be carried further within the same theory.

One can take this for a mere lack of data. Perhaps, so the natural expectation goes, deeper measurements or a deeper theory will suffice. But that expectation already assumes that better data will in the end also give better answers about origin.

The history of physics speaks cautiously against that hope. Newton brought celestial mechanics and falling bodies under a common law, but presupposed space, time, and mass. Maxwell bound electricity, magnetism, and light into a few equations, but presupposed charge and the continuum in which it appears. Einstein made gravitation into the geometry of spacetime, but took matter and coupling constants as given. Quantum field theory yields the most precise predictions we have, but in turn requires a symmetry group, a particle inventory, and constant parameters that it does not generate from itself.

The examples share a common form. Even the most successful theories explain more than their predecessors and yet still work with quantities, inventories, or constants they do not generate from themselves.

The point can also be made without that history. Data are always data of something: states, events, relations, measured values. For them to be data at all, something must be distinguished from something else. The beginning of the universe, however, is not simply one more state within an already differentiated world. It concerns the question of how such a world can come into force at all.

The first moment is therefore not only an unsolved problem of physics. It marks the place where physics touches its own presupposition. Physics works with distinguishable states. Remove that presupposition, and no measurement, no dynamics, no information, and no theory remain.

The question is therefore not only: which state was the first? But: what must hold for states to be in force at all? Then the issue is no longer only the universe, but the condition under which anything can appear as different at all. If states occur only as distinguished states, then information too depends on distinction. That raises the next question: why exactly?

Information

Information presupposes distinction. A measured value is this value only because it is not another. A recorded bit is this bit only because a different outcome remains excluded.

Spencer Brown places at the beginning of his calculus the drawn distinction. Wheeler reads physical reality as an answer to yes-or-no questions. In both cases information hangs on retained difference.

For the rest of the inquiry, that becomes a working assumption. Every piece of information rests on a successful act of distinction. *Successful* means: the difference must be stateable, it must not collapse at once, and it must be determinate enough to present itself again as the same.

The question thus shifts from a mere limit to the required performance. What must a distinction do if it is to carry information? Can a difference be stated at all? Does it hold? And is a field of such differences complete, with neither less nor more?

Chapter 2

Conditions of Stable Distinction

The previous question asks after a performance, not a boundary. The answer develops in three stages: a difference must be stateable, it must hold, and an entire field of such differences must stand exactly. No one stage is enough by itself. Only together do they yield the conditions under which a distinction can persist stably.

Distinguishability

What is the minimum without which a distinction does not succeed? Distinction is not a thing here but a performance: the holding-apart of one from another and the maintenance of that difference.

A mark on a page counts as a sign only because it stands out from the page — because something is read *as* mark and the rest *as* background. A measurement counts as a value only because it could have come out differently: this value and not that one. In both cases the issue is not yet counting or structure. It is the elementary possibility that something can appear as this rather than that.

That is the first demand. Before anything can be counted, ordered, or measured, a difference must be available. Call that distinguishability.

Nothing has yet been said about number, order, or stability. Distinguishability is only the availability of a contrast. But it does not suffice. A stated difference could collapse at once. What is needed so that it does not at once fall back into one?

Non-Collapse

What distinguishability leaves open is whether a difference holds as a difference at all. Something can lie apart for a moment and in the next fall back into one. Then the difference carries nothing. No mark would hold anything. No measured value could rest on it. No state could be asserted against another.

That is why a second demand follows distinguishability. A difference does not suffice merely because it flashes up once. It must be secured against collapse. Call this demand Non-Collapse.

Long before there are scales, observers, or instruments, it must be possible for a difference not to vanish at once. Measurement does not create that. Recording does not create it. Both already presuppose it.

The same demand appears in physics as well, for example in the black-hole information problem. The issue there is not first of matter or energy, but of whether differences once achieved can be lost without remainder. This is not a proof of the present argument, but it is a clear physical case of the same structure.

The demand can now be sharpened. Non-Collapse means: there must be a genuine separation between two positions, and that separation may not fall back into identity. That is exactly why the formal part will later have to work with inequality and equality.

With this much in hand, the next question arises. Is a single difference secured against collapse enough, or must an entire field of such differences stand complete?

Exact Plurality

The next question no longer concerns a single difference, but an entire field of differences. What does it take for a system to be fixed to a determinate plurality: that exactly these positions are here, neither fewer nor more?

This demand is felt most directly by trying to hold a small plurality in place and seeing what can go wrong. Consider a few blank sheets on a table and suppose there is a definite number of them. Two failures lie in wait, and they pull in opposite directions. The first: some of the sheets counted as different might not really be different at all — if every way of pointing to one points equally well to another, then there are not distinct sheets, but one sheet pointed at twice. Call this collapse: the many secretly fall together into fewer. The second, less obvious case: there may be more on the table than the count ever mentioned — a sheet at the edge that the accounting never ruled out. Call this surplus: the field hides elements the count never closed.

These two dangers are distinct. Collapse means fewer than claimed. Surplus means more than claimed. Whoever prevents only collapse may still have too much in the field. Whoever excludes only surplus may still secretly identify two positions. An exact plurality has to answer both dangers at once.

The demand that answers both has exactly two parts. The named positions must really lie apart. Call that *separation.* And every position of the field must be captured by the account. Call that *cover.* Separation prevents collapse. Cover prevents surplus. Only both together turn named positions into an exact plurality.

With separation and cover we have, for the first time, a specification complete enough to be written down exactly.

Ask where these two conditions can first be met completely, and the answer is: on the smallest field on which anything can be held apart at all, namely a field of two positions. With two, and not before, separation and cover can be written down completely and finitely. Two is not the origin of distinctness. Two is the first place at which its general conditions become fully explicit.

If this is the minimal performance stable information requires, how little must be granted to write it down exactly?

Chapter 3

The Threshold of Formalization

The previous chapter narrowed the question to a small finite specification: separation and cover on a field of two positions. At that point ordinary language reaches its limit. What is now missing is not another argument, but a tool in which writing and checking are one and the same act.

Up to this point, the text has unfolded philosophical motives and refined working assumptions. From here on it moves to the level of formal commitments and formal results. The question is no longer which orientation is plausible, but in what language the minimal conditions of a successful distinction can be written so exactly that they can actually be checked. Physical readings do not enter at this point.

What follows is not a tutorial in Agda. What matters is only why exactly these few means are needed. Each of them answers a question that has already arisen in the earlier movement.

There is such a language. The one used here is Martin-Löf type theory in the form realized by the proof assistant Agda. In it, to assert is to construct. A statement is a type, a proof is a term inhabiting that type, and a machine decides whether the term truly has the type claimed. A proof that does not check is here simply a file that does not compile. The argument is formulated in type theory not because it is the only possible carrier, but because it makes the cost of every step explicit.

Everything the argument uses has to be declared, and two settings, fixed at the head of the file, govern everything that follows. The flag `--safe` closes every escape hatch: no postulates, no principle asserted without proof, no back door through which an unearned result could enter. The flag `--without-K` withholds a convenient but additional assumption about equality, so that nothing in the development rests on more than the bare identity type. Under them, only the things we build exist, in full view.

What is granted can be said briefly: a carrier with two positions, a witness that the two are not equal, and an account that leaves no point of the carrier unaddressed. No numbers, no geometry, no physics, and no classical logic beyond what is constructed by hand.

If the formalization later seems to do more than this brief list suggests, that is only because much already follows from little. It is not because more was smuggled in beforehand.

What follows is the same thought under stricter discipline. The previous chapters named in ordinary language the performances a stable distinction must carry. The file now writes those performances down one by one in a language where hidden shortcuts are not allowed.

The order of writing is not the same as the order of the inquiry. In the inquiry, distinction, Non-Collapse, and exact plurality came first. In the file, what comes first is what the checker needs in order for those conditions to be sayable at all. No new content is introduced here. The file simply makes available the few means without which separation and cover could not be stated exactly. It begins with the mere bookkeeping of universe levels. Nothing mathematical is imported with that.

```
open import Agda.Primitive using (Level; lzero; lsuc; _⊔_; Setω)
```

That first line is only housekeeping. It gives the checker a way to track sizes consistently, and none of the conceptual weight of the argument rests on it. The real formal work begins with the thinnest device needed to say when something is the same and when it is not.

That is why equality comes first in the file. Not because sameness would be prior to distinction in the order of the book, but because separation and cover are not even formulable without a minimal way to say same, not the same, and substitution. The identity type gives exactly that. At this stage it is not meant to do more.

```
infix 4 _≡_

data _≡_ {ℓ : Level} {A : Set ℓ} (x : A) : A → Set ℓ where
 refl : x ≡ x
{-# BUILTIN EQUALITY _≡_ #-}
```

```
sym : {ℓ : Level} {A : Set ℓ} {x y : A} → x ≡ y → y ≡ x
sym refl = refl

trans : {ℓ : Level} {A : Set ℓ} {x y z : A} → x ≡ y → y ≡ z → x ≡ z
trans refl q = q

cong : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} {B : Set ℓ₂} (f : A → B) {x y : A} → x ≡ y → f x ≡ f y
cong f refl = refl

subst : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} (P : A → Set ℓ₂) {x y : A} → x ≡ y → P x → P y
subst P refl px = px
```

Next the file needs a way to say impossibility. A guard against collapse is worthless unless one can express that an identification is excluded. The empty type gives that means. Negation is then the map into impossibility, and inequality — the formal shape of being kept apart — is the negation of equality.

```
data ⊥ : Set where

⊥-elim : {ℓ : Level} {A : Set ℓ} → ⊥ → A
⊥-elim ()

¬_ : {ℓ : Level} → Set ℓ → Set ℓ
¬ A = A → ⊥

infix 2 _≢_
_≢_ : {ℓ : Level} {A : Set ℓ} → A → A → Set ℓ
x ≢ y = ¬ (x ≡ y)
```

Then the file needs a way to say exactly: one of two, and which one. Cover cannot be expressed by an untagged choice, because that would merge the two cases. The disjoint union keeps the mark, so that a later case analysis can say from which side a witness comes.

```
infixr 2 _⊎_

data _⊎_ {ℓ : Level} (A B : Set ℓ) : Set ℓ where
 inj₁ : A → A ⊎ B
 inj₂ : B → A ⊎ B
```

Finally, the file needs a way to carry a thing together with a statement about it. That is what the dependent pair does, with the ordinary product as its constant special case. In this way an existential claim becomes an actual witness together with a proof about it.

```
infixr 3 _×_

record _×_ (A B : Set) : Set where
 constructor _,_
```

```
  field
    fst : A
    snd : B

open _×_ public

record Σ (A : Set) (B : A → Set) : Set where
  constructor _,_
  field
    fst : A
    snd : B fst

open Σ public
```

With these means in place, the earlier question can for the first time be translated into a single exact object. What does this smallest object look like, the one that carries separation and cover together?

Chapter 4

The Distinction Record

The previous question asked how little has to be granted for the minimal conditions of successful distinction to be writable exactly. The Distinction Record is the first complete answer. Formally it is a record. In content it is not an inventory of a thing, but a certificate of conditions: exactly the data needed for a difference to persist stably as a difference.

```
record Distinction : Set1 where
 field
  S    : Set
  ℓ    : S
  r    : S
  ℓ≠r : ℓ ≠ r
  cover : (x : S) → (x ≡ ℓ) ⊎ (x ≡ r)

open Distinction public
```

Read back into prose, `S` is the carrier, ℓ and `r` are the two positions, ℓ≠`r` is separation, and `cover` is cover. The record contains no more than that. That is its point. It does not primarily describe an object. It describes the minimal configuration under which a distinction can be held fast as a successful distinction. Its first consequence is simple: `cover` says that every point of the carrier reduces to ℓ or `r`.

```
law1-1-cover : (d : Distinction) → (x : S d) → (x ≡ ℓ d) ⊎ (x ≡ r d)
law1-1-cover = cover
```

If every point is one of the two, then it is enough to check a property at ℓ and `r` in order to obtain it for all points. That is the carrier eliminator.

```
Distinction-elim :
 (d : Distinction) →
```

```
{P : S d → Set} →
P (ℓ d) →
P (r d) →
(x : S d) →
P x
Distinction-elim d {P} pℓ pr x with cover d x
... | inj₁ x≡ℓ = subst P (sym x≡ℓ) pℓ
... | inj₂ x≡r = subst P (sym x≡r) pr
```

Surplus is therefore not only excluded but eliminated by construction. The fourth field does the opposite work. ℓ≠r forbids the collapse of the two positions into one. Anyone who tries to identify them produces a contradiction the checker rejects.

Taken together, separation and cover yield a stable two. The next question is what self-maps such a carrier admits.

Chapter 5

What Already Follows from the Record

Once the Distinction Record is given, the next question follows at once: what self-maps does such a carrier admit? Once the carrier is fixed to two classes, self-maps are almost entirely determined by their values on ℓ and `r`. To compare two maps, we use pointwise equality. On this carrier it is especially manageable because `cover` reduces every argument to one of the two boundary cases.

```
infix 4 _≗_
_≗_ : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} {B : Set ℓ₂} → (A → B) → (A → B) → Set (ℓ₁ ⊔ ℓ₂)
_≗_ {A = A} f g = (x : A) → f x ≡ g x

id : {A : Set} → A → A
id x = x
```

Among these maps one is especially important: the map that exchanges the two positions, sending ℓ to `r` and `r` to ℓ. Call it the dual. Applied twice, it returns every point to its starting place.

```
Distinction-dual : (d : Distinction) → S d → S d
Distinction-dual d x with cover d x
... | inj₁ _ = r d
... | inj₂ _ = ℓ d

Distinction-dual-involutive :
 (d : Distinction) →
 (x : S d) →
 Distinction-dual d (Distinction-dual d x) ≡ x
Distinction-dual-involutive d =
 Distinction-elim d proof-ℓ proof-r
 where
  proof-ℓ : Distinction-dual d (Distinction-dual d (ℓ d)) ≡ ℓ d
```

```
proof-ℓ with cover d (ℓ d)
... | inj₂ ℓ≡r = ⊥-elim ((ℓ≢r d) ℓ≡r)
... | inj₁ _ with cover d (r d)
... | inj₁ r≡ℓ = ⊥-elim ((ℓ≢r d) (sym r≡ℓ))
... | inj₂ _ = refl

proof-r : Distinction-dual d (Distinction-dual d (r d)) ≡ r d
proof-r with cover d (r d)
... | inj₁ r≡ℓ = ⊥-elim ((ℓ≢r d) (sym r≡ℓ))
... | inj₂ _ with cover d (ℓ d)
... | inj₂ ℓ≡r = ⊥-elim ((ℓ≢r d) ℓ≡r)
... | inj₁ _ = refl
```

The central question is: how many maps are there, counted up to pointwise equality? The boundary table allows exactly four combinations, one for each pair of choices at ℓ and `r` — the constant-left map, the constant-right map, the identity, and the dual. We name those four cases explicitly.

```
data EndoCase : Set where
 case-constL : EndoCase
 case-constR : EndoCase
 case-id     : EndoCase
 case-dual : EndoCase
```

The four names must now be tied to actual maps. Inside a module parametrized by a fixed distinction, `interpret` turns each label into its endomorphism, and `classify` reads the matching label back from an endomorphism's two boundary values.

```
module K₄ (d : Distinction) where
 Endo : Set
 Endo = S d → S d

 ≗-refl : {f : Endo} → f ≗ f
 ≗-refl x = refl

 ≗-sym : {f g : Endo} → f ≗ g → g ≗ f
 ≗-sym p x = sym (p x)

 ≗-trans : {f g h : Endo} → f ≗ g → g ≗ h → f ≗ h
 ≗-trans p q x = trans (p x) (q x)

 constL : Endo
 constL _ = ℓ d

 constR : Endo
 constR _ = r d

 dual : Endo
 dual = Distinction-dual d
```

```
dual-ℓ : dual (ℓ d) ≡ r d
dual-ℓ with cover d (ℓ d)
... | inj₁ _ = refl
... | inj₂ ℓ≡r = ⊥-elim ((ℓ≢r d) ℓ≡r)

dual-r : dual (r d) ≡ ℓ d
dual-r with cover d (r d)
... | inj₁ r≡ℓ = ⊥-elim ((ℓ≢r d) (sym r≡ℓ))
... | inj₂ _ = refl

interpret : EndoCase → Endo
interpret case-constL = constL
interpret case-constR = constR
interpret case-id     = id
interpret case-dual = dual

classify : Endo → EndoCase
classify f with cover d (f (ℓ d)) | cover d (f (r d))
... | inj₁ _ | inj₁ _ = case-constL
... | inj₂ _ | inj₂ _ = case-constR
... | inj₁ _ | inj₂ _ = case-id
... | inj₂ _ | inj₁ _ = case-dual
```

Reading a function's label and then interpreting it returns the same function: at the two boundary points by direct inspection, and therefore everywhere by the eliminator. The two boundary lemmas this rests on are routine case analyses; they are part of the checked file and are not reprinted here.

```
classify-sound : (f : Endo) → interpret (classify f) ≗ f
classify-sound f x = Distinction-elim d (sound-at-ℓ f) (sound-at-r f) x
```

The label is not merely available; it is forced. No two distinct labels can interpret to the same function — sixteen comparisons, one for each ordered pair of labels, every one discharged by the single separation witness ℓ≢r. That case analysis is routine and lives in the checked file without crowding the page. From it, classification is unique, and reading a label out and back in returns the very same label.

```
classify-unique : (f : Endo) → (c : EndoCase) → interpret c ≗ f → c ≡ classify f
classify-unique f c c≗f =
 interpret-injective c (classify f) (≗-trans c≗f (≗-sym (classify-sound f)))

classify-interpret : (c : EndoCase) → classify (interpret c) ≡ c
classify-interpret c = sym (classify-unique (interpret c) c ≗-refl)
```

Together this says: the four labels are pairwise distinct, and every self-map falls uniquely under one of them.

```
EndoCase-distinct :
 (case-constL ≡ case-constR → ⊥) ×
```

```
 (case-constL ≡ case-id    → ⊥) ×
 (case-constL ≡ case-dual → ⊥) ×
 (case-constR ≡ case-id    → ⊥) ×
 (case-constR ≡ case-dual → ⊥) ×
 (case-id ≡ case-dual       → ⊥)
EndoCase-distinct =
 (λ ()) ,
 ((λ ()) ,
  ((λ ()) ,
   ((λ ()) ,
    ((λ ()) ,
     (λ ())))))
```

Gathered into two top-level statements, the result reads cleanly: every endomorphism of a distinction is one of the four cases (soundly), and is so in exactly one way (uniquely).

```
k4-classification-sound :
 (d : Distinction) →
 (f : S d → S d) →
 Σ EndoCase (λ c → K₄.interpret d c ≗ f)
k4-classification-sound d f = K₄.classify d f , K₄.classify-sound d f

k4-classification-unique :
 (d : Distinction) →
 (f : S d → S d) →
 (c₁ c₂ : EndoCase) →
 K₄.interpret d c₁ ≗ f →
 K₄.interpret d c₂ ≗ f →
 c₁ ≡ c₂
k4-classification-unique d f c₁ c₂ p₁ p₂ =
 K₄.interpret-injective d c₁ c₂ (K₄.≗-trans d p₁ (K₄.≗-sym d p₂))
```

A quick algebraic over-reading is tempting. But what has been shown is only this: the four maps are closed under composition, and they do not form a group because the two constants are not invertible. What is established is the number of cases and their unique classification up to pointwise equality. Any richer algebraic reading — and likewise the later reading of this four-case structure as a K_4 closure — is an additional step.

The next answer has therefore been reached. From a separated and covered binary distinction follows a unique four-case classification of its self-maps. The next question no longer concerns the structure itself, but its status: what has thereby been shown, and what not?

Chapter 6

Result and Boundary

What has been shown is a small but precise stock: a carrier with two positions, a witness that they are apart, an exhaustion principle with no third case, and exactly four self-maps up to pointwise equality. Under `--safe` and `--without-K`, these statements are theorems in the strict sense: if any one of them failed, the file would not compile.

At this point the levels of the book can be ordered cleanly. Philosophical orientation was the conjecture that origin, determination, and information all point toward distinction. The working assumption was that successful distinction must satisfy minimal conditions. The formal result is the stock the file actually carries. It is on that level that this chapter moves.

What has not been shown is that the world can be derived from nothing, or that every determination necessarily presupposes distinction. That pre-formal thesis motivates the inquiry; it is not the inquiry's formal result. Nor does Agda prove that no other formal carrier could realize the same content. What has been shown is only that this content can be constructed here without additional assumptions.

The result is nonetheless interesting. Once two positions are granted in full view and kept apart, that minimal grant already carries a closed shape. This is the first point at which the conditions of stable distinction become exact and machine-checkable. The step from philosophical question to formal structure remains small. That is exactly why it is robust.

From here one can say exactly how far the step carries and where its boundary lies. Three different courts decide that: the compiler over the formal kernel, ordinary mathematics over the derivations around it, and experiment over any reading of the kernel as physics. That separation is not caution but a condition of honesty. The distance between a checked theorem and a measured quantity is the point at which a project can lose its integrity.

At the level of this book, the status is clear. That a stable distinction has exactly four self-maps up to pointwise equality is theorem. If one reads the four-case closure as a complete figure, then the number of its pairings, $\binom{4}{2} = 6$, is again pure arithmetic. None of that is a physical claim. They are integers, and they are theorems about integers.

The larger books continue that arithmetic. A polynomial in the closure's invariants, for instance, yields 137. As arithmetic, that too is theorem. Only when one reads such an integer as a constant of nature does theorem give way to structural hypothesis, and only measurement can decide the comparison. If a measurement fails, the identification fails; the arithmetic remains what it was.

The question of origin therefore remains unanswered. What has shifted is narrower and more precise: its first minimal step — the holding-apart of one position from another — is no longer mere intuition. It is present as normal form, four-case ledger, and return to its own datum, and a machine can check it.

Chapter 7

Outlook: Void and Form

This book stops at the first formally secured datum. *Void* asks what happens if that datum is not merely described but continued as a process. The question there is no longer what minimal structure a distinction must carry, but what follows mechanically from a separated and covered distinction under the same disciplines.

The first part of that continuation is purely formal. A binary carrier with separation and cover normalizes to `Two`. Its endomorphism space has exactly four cases. Under the stated representation conditions that four-case space closes to K_4, and the closure returns to the original `Distinction` datum. From there the development continues without new postulates: arithmetic is built internally, and where rational data no longer suffice, the continuous arena is constructed within the same development.

This yields a register that is initially purely mathematical. It contains finite invariants, arithmetic expressions, and the structures that follow from the closure. Only once this register is fully in place does the next question arise: whether its entries admit a physical reading.

That is where *Form* begins. *Form* takes the register produced in *Void* as a fixed formal constraint and tests sector assignments, correction chains, ratios, and mass readouts against it. The separation of levels remains strict. What the kernel forces remains theorem. The assignment of physical names to those structures is interpretation. Whether the interpretation holds is decided only by comparison with the world.

The continuation is therefore neither a new beginning nor a direct leap to physics. The process distinction is allowed to keep running under the same rules; its closure yields first normal form, four-case structure, K_4, register, and continuous arena. Only after that does the question arise whether this register can be read physically. This companion marks only the point at which that continuation begins.